PRE-FREEZE / DO NOT FABRICATE

CORE-S3 research schematic — ESP32-S3-WROOM-1-N16R8 (USB FS, 8 MB PSRAM). Do not order boards before P1 benchmark and spec/90 sign-off.
GPIO conflicts: 0 (check_pinmap.py). Docs: README.md, pinmap.md, power-tree.md, interfaces.md, comparison in README, PRE-FREEZE.

POWER: USB-C + BQ24074 charge/path, TPS63000 3V3 buck-boost (D-41), +-12V/+5V from PWR-2 J17, BL boost

MCU: ESP32-S3-WROOM-1-N16R8 (USB FS, 8 MB PSRAM), microSD SPI, BOOT/RESET, UART0 TP

AUDIO: PCM5102A DAC + PCM1808 ADC (I2S), TL072 LINE/PHONES, rear jacks

MIDI TRS-A: 6N137 IN / 74HCT125 OUT; GATE_OUT driver

CV: 2x MCP4728 (0x60/0x61), 6x TL074 x4.9, CV IN level shift (1-4), MCP3208 + chains

PANEL-2 rev.2 canon: J14 IDC-26 analog, J15 FFC-40 SPI display, J16 IDC-16 digital; rear jacks; J17/J18 PWR-2

TEST POINTS (TP1..TP22)

POWER FLAGS (net drivers for ERC)

Built on open hardware - CERN-OHL-S-2.0

Title: CORE-S3 PRE-FREEZE — ESP32-S3 (not for fab)
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